SAMPLE

Great Observatories: Square Kilometre Array

The Future of Radio Astronomy

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Introduction

The story begins in two places at once, and that doubling is the point. Eight hundred kilometers north of Perth, on Wajarri country in the red dirt of Western Australia, a forest of metal antennas rises from a plain so quiet that a mobile phone is banned from the site. Six thousand kilometers away, in the Karoo of South Africa's Northern Cape, on land where sheep outnumber people by an enormous margin, dishes the size of town water tanks turn on their mounts toward the Milky Way. Both places belong to one instrument, the Square Kilometre Array, the largest telescope project human beings have ever organized, and the account that follows keeps both sites in view because the telescope itself keeps both in view.

The name states the founding ambition plainly: a collecting surface of one million square meters, a full square kilometre of antenna, aimed at the sky. No such machine has existed. The generation of radio telescopes before it worked with dishes and arrays whose areas ran to some tens of thousands of square meters, and the leap in scale demanded new engineering, new mathematics, new computing, and a new kind of international organization to pay for it. The telescope is not one instrument but two coordinated arrays, a low frequency field of simple antennas in Australia and a mid frequency field of dishes in South Africa, run as a single observatory from a headquarters in the English countryside, and by the end of this volume the reader will hold the whole arrangement in mind, from the first dipole to the exabyte stream.

The plan of the account follows the series convention. Early chapters give the history of the idea and the machine, the middle chapters the science and the engineering, the later chapters the discoveries the telescope is built to chase and the honest difficulties that stand in the way, and the closing chapters the pocket delights, the verdict, and the shelf. The Square Kilometre Array deserves the full treatment for a reason the introduction will state once and return to often: it is the first telescope on this scale designed around the universe's most abundant signal, the faint hiss of neutral hydrogen, and the machines built to hear that hiss across thirteen billion years of travel time will also time the pulses of dead stars, weigh the gas of unborn galaxies, and listen, whether anyone admits it or not, for anyone broadcasting.

The account opens with a paragraph of honesty, since the subject is an instrument still under construction and the reader deserves to know what kind of story this is. Some chapters describe finished machines and measured results. Others describe the best current plans for a telescope whose arrays are still being installed, whose first campaigns have only begun, and whose designers themselves publish careful statements of what they expect rather than what they know. The volume keeps the two registers distinct, marks the boundary between the measured and the hoped for, and lets the machine stand as the largest construction site in the history of astronomy, the loudest quiet place on Earth, and a bet, made by fourteen governments on two continents, that the sky will be worth the money.

A note on the numbers this book uses, because a telescope described by its statistics can drown a reader in them. Where the account quotes a count, a frequency, or a data rate, the figure is the design's declared value from the project's own published plan, and the book keeps to the round, stable ones: the antenna counts, the site separations, the survey speeds. Where the engineering is still moving, and on a machine this young it often is, the text says so and gives the shape of the change rather than a decimal that will be wrong by the time the ink dries; the reader who wants the living numbers will find them in the observatory's own publications, and the reader who wants the machine itself will find it here, in its plan and its promise, which is what a book can hold honestly.

A note on the reader's likely questions, because a book about a telescope that is mostly mathematics has to answer the layman's first ones plainly. The machine cannot see planets or photograph faces; it hears, in the radio, and its pictures are maps of signals, translated by the pipelines into the colors of the human eye. The machine is not finished, and this book is honest about what exists, what is building, and what is planned, three states the text distinguishes in every chapter; and the machine belongs to its member nations, so the reader's own country may hold a share of it, the taxes of many hands paying for the antennas' arithmetic. The reader who finishes the volume will be able to read the project's news with understanding, the account's own success test.

Chapter 1: The Discovery of Square Kilometre Array

No single person discovered the Square Kilometre Array, and the chapter's title uses the series convention honestly rather than literally: what was discovered, across the 1990s, was that a machine of this scale had become thinkable. Radio astronomers had wanted more collecting area since the field's earliest years, when the war's

surplus radar dishes were turned toward the sky and the first maps of the radio heavens were drawn by hand. The founding calculation was simple and brutal. Every doubling of collecting area halves the time needed to detect a faint source and doubles the volume of space a survey can reach, and by the 1980s the field had enough sensitivity to see the obvious: the next real advance in radio astronomy would require an antenna measured in square kilometres, a phrase so large it sounded like a misprint.

The idea acquired an institutional shape at the start of the 1990s, when an international working group began meeting under radio astronomy's scientific unions to ask what such an instrument would need and what it would cost. The name settled early and never changed. A working name that is simply a number has a way of focusing a community, and the "square kilometre array" became the field's shared long term goal, cited in strategy papers and grant proposals for two decades before a single production antenna was ordered. The first years produced a design problem with no obvious answer, since a single dish with a million square meters of surface would be a bowl kilometers wide, an engineering absurdity, and the field instead had to invent its way toward arrays of cheap antennas whose signals could be combined in software.

The 1990s and 2000s were therefore the era of pathfinders. Collaborations around the world built small arrays to test the technologies a full machine would demand: antennas cheap enough to be manufactured by the thousand, receivers stable enough to combine signals a hundred kilometers apart, and computing farms powerful enough to do the arithmetic of correlation. Each pathfinder answered one question and raised two more, and the design converging through those years was genuinely new, an instrument that would spend most of its budget not on steel and concrete but on fiber, processors, and the software that turns a field of antennas into a single coherent eye.

The site question took a decade of its own. A telescope that listens for signals fainter than a phone's leakage from another continent cannot sit near any city, and candidate regions in Argentina, Australia, China, and South Africa were surveyed through the late 2000s for their radio quietness, their skies, their land use, and their politics. The decision, announced in 2012, was the rare compromise that actually made scientific sense: the array would be split, with the low frequency antennas going to the radio quiet Shire of Murchison in Western Australia and the mid frequency dishes going to the Karoo, since the two designs needed different sites and the two hemispheres gave the single observatory the whole sky between them.

The competition's inside history deserves one honest paragraph, because the episode is the project's most human chapter. Four candidate regions entered the final decade of assessment: arid flats in Argentina, sites in inland China, the Shire of Murchison in Australia, and the Karoo in South Africa. Two were eliminated early on technical grounds, and the final pair, Australia and South Africa, each offered such strong conditions that the assessors found no clean loser. The winner's announcement, delivered at a meeting in the Netherlands in 2012, awarded the low frequency field to Australia and a share of the mid frequency work to both finalists, a split the community initially read as diplomatic softening and later came to read as engineering honesty: the two designs genuinely did need different landscapes. The decision's aftermath was gracious, and within a year both host countries had written the telescope into their national science strategies and their law.

The organizational machinery took a decade longer. The Square Kilometre Array Organisation, founded in 2011 to hold the design work, was succeeded by a formal intergovernmental organization, established by treaty in 2021 as one of the larger science undertakings ever set up under international law, with headquarters at the Jodrell Bank Observatory outside Manchester. Construction at both sites began with ceremonies in December 2022, thirty years after the working groups first met.

The member roster that stood behind the treaty reflects the project's decades of recruitment. The founding group counted Australia and the United Kingdom, China, Italy, the Netherlands, Portugal, Switzerland, and the host nation South Africa among its members, with further states joining or negotiating across the years that followed, and several major radio astronomy nations holding associate status while their accession proceeded. The chapter's history is therefore also a geography lesson: a telescope conceived by a working group, designed by a consortium, sited by a competition, and chartered by a treaty is an instrument whose parts list includes parliaments. The chapter's discovery, in the end, is that the machine took longer to organize than the transcontinental railways, and that the patient part, the diplomacy, was as hard as any antenna.

The working-group decade deserves its paragraph, because the machine's true origin is a set of committees, and the account should say so with a straight face. Through the nineteen-nineties the radio unions ran a standing question: what would the next-generation instrument need to be, if radio astronomy were to keep pace with the optical giants being planned for the same era; the answers converged from several directions, a collecting area of a square kilometre, the namesake figure, priced far beyond any single nation's science budget, and the decade's

papers and workshops built the case that only a global consortium could build it. The discovery, in the chapter's honest sense, was thus institutional: the astronomers discovered that their next telescope had to be an observatory of nations, and the consortium they sketched in the nineties is the organization that exists today.

The site competition deserves its own paragraph, because the two-site outcome was the longest drama in the project's middle age. The bidding nations, among them Australia, China, South Africa, and Argentina, offered land, quiet skies, and national investment; the evaluations ran for years, measuring radio silence, atmosphere, infrastructure, and the host countries' commitments; and the decision, when it came in 2012, was the compromise the project's dual design made possible, the low-frequency array to Australia and the mid-frequency array to Southern Africa, with both sites preserving the planet's two great radio-quiet reserves. The chapter keeps the drama's lesson: the machine's geography was chosen the way all great observatory sites are chosen, by the physics of quietness first and the politics of partnership second.